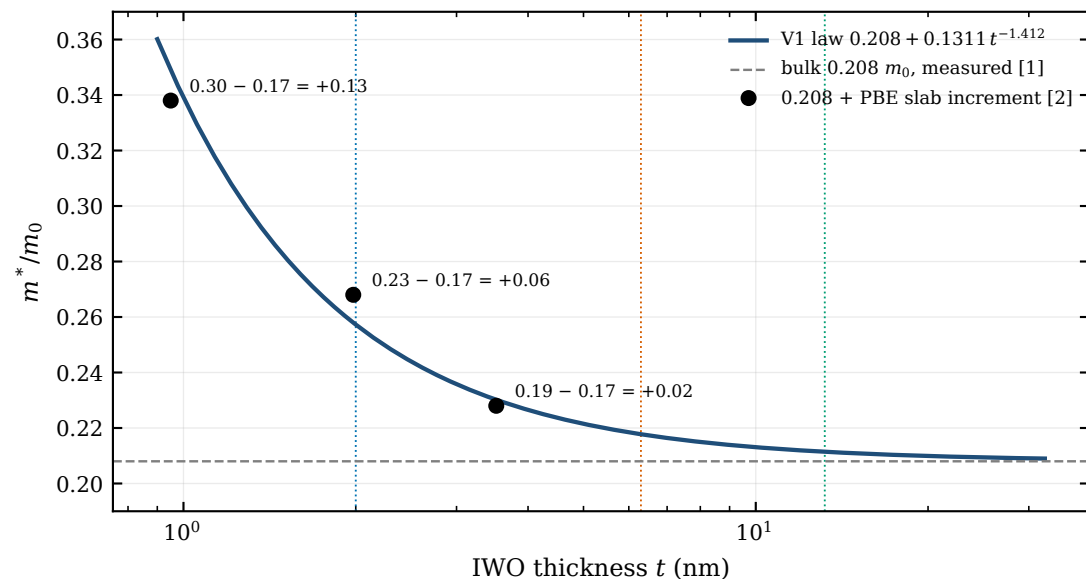


(b) Electron effective mass $m^*(t)$, model V1



What is plotted

Effective mass of the conduction electrons. The bulk value is a measurement; only the change with thickness is taken from DFT: each dot is $0.208 m_0$ plus the PBE slab mass minus the PBE bulk mass of [2] (written next to each dot). The curve is a log-log fit through these three increments (scripts/material_laws.py). In ATLAS the mass enters only through the conduction-band density of states, $N_c \propto (m^*)^{3/2}$.

References

[1] Stokey et al., J. Appl. Phys. 129, 225102 (2021): $0.208 m_0$ from optical Hall measurements on an In_2O_3 single crystal ($n = 2.8 \times 10^{17} \text{ cm}^{-3}$), Abstract and Sec. IV.C. [2] Lin et al., ACS Nano 16, 21536 (2022), p. 21540: PBE masses $0.30 / 0.23 / 0.19 m_0$ for $0.95 / 1.98 / 3.52 \text{ nm}$ slabs and $0.17 m_0$ for bulk.